ROCSCIENCE

Case4
Slide2 - An Interactive Slope Stability Program
Date Created: 13/03/2026, 8:34:42 am
Software Version: 9.041

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Slide2 Analysis Information

Case4

Project Summary

File Name:	Case4.slmd
Slide2 Modeler Version:	9.041
Compute Time:	00h:00m:05.906s
Project Title:	Slide2 - An Interactive Slope Stability Program
Date Created:	13/03/2026, 8:34:42 am

General Settings

Units of Measurement:	Metric Units
Time Units:	days
Permeability Units:	meters/second
Data Output:	Maximum
Failure Direction:	Right to Left

Analysis Options

Slices Type:	Vertical
Analysis Methods Used	
	Bishop simplified
Number of slices:	25
Tolerance:	0.0001
Maximum number of iterations:	100
Check malpha < 0.2:	Yes
Create Interslice boundaries at intersections with water tables and piezos:	Yes
Initial trial value of FS:	1
Steffensen Iteration:	Yes

Groundwater Analysis

Groundwater Method:	Water Surfaces
Pore Fluid Unit Weight [kN/m3]:	9.81
Advanced Groundwater Method:	None

Random Numbers

Pseudo-random Seed:	10116
Random Number Generation Method:	Park and Miller v.3

Surface Options

Surface Type:	Circular
Search Method:	Auto Refine Search
Divisions along slope:	30
Circles per division:	15
Number of iterations:	15
Divisions to use in next iteration:	50%
Composite Surfaces:	Disabled
Minimum Elevation:	Not Defined
Minimum Depth:	Not Defined
Minimum Area:	Not Defined
Minimum Weight:	Not Defined

Seismic Loading

Advanced seismic analysis:	No
Staged pseudostatic analysis:	No

Materials

Material 1	
Color	
Strength Type	Mohr-Coulomb
Unit Weight	16 kN/m3
Cohesion	9 kPa
Phi	32 °
Water Surface	None
Ru Value	0

Global Minimums

Method: bishop simplified

FS	1.234670
Center:	21.024, 67.292
Radius:	43.234
Left Slip Surface Endpoint:	30.000, 25.000
Right Slip Surface Endpoint:	58.068, 45.000
Resisting Moment:	55184.5 kN-m
Driving Moment:	44695.6 kN-m
Total Slice Area:	113.545 m2
Surface Horizontal Width:	28.0675 m
Surface Average Height:	4.04543 m

Global Minimum Support Data

No Supports Present

Valid and Invalid Surfaces

Method: bishop simplified

Number of Valid Surfaces:	38360
Number of Invalid Surfaces:	0

Slice Data

Global Minimum Query (bishop simplified) - Safety Factor: 1.23467

Slice Number	Width [m]	Weight [kN]	Angle of Slice Base [deg]	Base Material	Base Cohesion [kPa]	Base Friction Angle [deg]	Shear Stress [kPa]	Shear Strength [kPa]	Base Normal Stress [kPa]	Pore Pressure [kPa]	Effective Normal Stress [kPa]	Base Vertical Stress [kPa]	Effective Vertical Stress [kPa]
1	1.1227	5.78597	12.746	Material 1	9	32	8.88095	10.965	3.14472	0	3.14472	5.15362	5.15362
2	1.1227	17.073	14.2763	Material 1	9	32	13.276	16.3915	11.829	0	11.829	15.2071	15.2071
3	1.1227	27.7844	15.8172	Material 1	9	32	17.3297	21.3964	19.8384	0	19.8384	24.7478	24.7478
4	1.1227	37.9074	17.3699	Material 1	9	32	21.0459	25.9847	27.1813	0	27.1813	33.7645	33.7645
5	1.1227	47.4276	18.9359	Material 1	9	32	24.4278	30.1603	33.8636	0	33.8636	42.2442	42.2442
6	1.1227	56.3285	20.5168	Material 1	9	32	27.4776	33.9258	39.8897	0	39.8897	50.1723	50.1723
7	1.1227	64.5915	22.1141	Material 1	9	32	30.1966	37.2828	45.262	0	45.262	57.5322	57.5322
8	1.1227	72.1952	23.7298	Material 1	9	32	32.5849	40.2316	49.981	0	49.981	64.305	64.305
9	1.1227	79.1157	25.3658	Material 1	9	32	34.6418	42.7712	54.0453	0	54.0453	70.4691	70.4691
10	1.1227	85.3256	27.0243	Material 1	9	32	36.3658	44.8998	57.4517	0	57.4517	76.0004	76.0004
11	1.1227	90.7938	28.7076	Material 1	9	32	37.7541	46.6138	60.1947	0	60.1947	80.8709	80.8709
12	1.1227	95.4849	30.4185	Material 1	9	32	38.8028	47.9087	62.2669	0	62.2669	85.0493	85.0493
13	1.1227	99.3581	32.16	Material 1	9	32	39.5072	48.7784	63.6587	0	63.6587	88.4992	88.4992
14	1.1227	102.367	33.9355	Material 1	9	32	39.861	49.2152	64.3577	0	64.3577	91.179	91.179
15	1.1227	104.457	35.7489	Material 1	9	32	39.8566	49.2098	64.3491	0	64.3491	93.0406	93.0406
16	1.1227	105.565	37.6046	Material 1	9	32	39.4851	48.7511	63.615	0	63.615	94.0277	94.0277
17	1.1227	105.617	39.5079	Material 1	9	32	38.7356	47.8257	62.1342	0	62.1342	94.0744	94.0744
18	1.1227	104.526	41.465	Material 1	9	32	37.5956	46.4182	59.8817	0	59.8817	93.1025	93.1025
19	1.1227	102.186	43.4831	Material 1	9	32	36.0503	44.5102	56.8281	0	56.8281	91.0184	91.0184
20	1.1227	98.4701	45.5713	Material 1	9	32	34.0822	42.0803	52.9396	0	52.9396	87.7083	87.7083
21	1.1227	93.2197	47.7403	Material 1	9	32	31.6714	39.1037	48.176	0	48.176	83.0317	83.0317
22	1.1227	86.237	50.0041	Material 1	9	32	28.7943	35.5515	42.4913	0	42.4913	76.8121	76.8121
23	1.1227	72.9413	52.3805	Material 1	9	32	24.247	29.9371	33.5063	0	33.5063	64.9696	64.9696
24	1.1227	46.0909	54.893	Material 1	9	32	16.3186	20.1481	17.8406	0	17.8406	41.0536	41.0536
25	1.1227	15.8735	57.5741	Material 1	9	32	8.03978	9.92647	1.48267	0	1.48267	14.1387	14.1387

Interslice Data

Global Minimum Query (bishop simplified) - Safety Factor: 1.23467

Slice Number	X coordinate [m]	Y coordinate - Bottom [m]	Interslice Normal Force [kN]	Interslice Shear Force [kN]	Interslice Force Angle [deg]
1	30	25	0	0	0
2	31.1227	25.254	9.17198	0	0
3	32.2454	25.5396	20.6977	0	0
4	33.3681	25.8577	33.8439	0	0
5	34.4908	26.2089	47.9264	0	0
6	35.6135	26.5941	62.3081	0	0
7	36.7362	27.0142	76.3982	0	0
8	37.8589	27.4704	89.6511	0	0
9	38.9816	27.9639	101.567	0	0
10	40.1043	28.4962	111.692	0	0
11	41.227	29.0688	119.621	0	0
12	42.3497	29.6837	124.996	0	0
13	43.4724	30.3429	127.515	0	0
14	44.5951	31.0488	126.933	0	0
15	45.7178	31.8042	123.066	0	0
16	46.8405	32.6124	115.807	0	0
17	47.9632	33.4772	105.126	0	0
18	49.0859	34.4029	91.0938	0	0
19	50.2086	35.395	73.8961	0	0
20	51.3313	36.4597	53.8603	0	0
21	52.454	37.6051	31.4919	0	0
22	53.5767	38.8406	7.52393	0	0
23	54.6994	40.1788	-17.0098	0	0
24	55.8221	41.6356	-38.6008	0	0
25	56.9448	43.2327	-48.772	0	0
26	58.0675	45	0	0	0

Discharge Sections

Entity Information

Group 1

Shared Entities

Type	Coordinates (x,y)
External Boundary	95, 20 95, 45 55, 45 30, 25 10, 25 10, 20